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EDUCATION

- **Ph.D. in Computer Science and Engineering** Apr. 2021 – Mar. 2024
Department of Computer Science and Communications Engineering, Waseda University, Japan
 - Supervisor: Tetsunori Kobayashi, Tetsuji Ogawa
 - Thesis title: “Incorporating Linguistic Cues for End-to-End Speech Recognition”
- **M.E. in Computer Science and Engineering** Apr. 2019 – Mar. 2021
Department of Computer Science and Communications Engineering, Waseda University, Japan
 - Supervisor: Tetsunori Kobayashi, Tetsuji Ogawa
 - Thesis title: “Non-Autoregressive End-to-End Speech Recognition with CTC and Mask-predict”
- **B.E. in Computer Science and Engineering** Apr. 2015 – Mar. 2019
Department of Computer Science and Engineering, Waseda University, Japan
 - And with a minor in Intermedia Art and Science
 - Supervisor: Tetsunori Kobayashi, Tetsuji Ogawa
 - Thesis title: “Speaker Invariant Feature Extraction for Zero-Resource Languages”

EXPERIENCE

- **Google LLC** Oct. 2022 – Jan. 2023
Research Intern New York, USA
 - Mentor: Andrew Rosenberg, Bhuvana Ramabhadran
 - Worked on text injection in speech recognition
 - Major publication(s): [Higuchi+, ASRU’23]
- **Carnegie Mellon University** Mar. 2022 – June 2022
Visiting Scholar Pennsylvania, USA
 - Mentor: Shinji Watanabe
 - Worked on language modeling in speech recognition
 - Major publication(s): [Higuchi+, EMNLP Findings’22] [Higuchi+, ICASSP’23b]
- **Mitsubishi Electric Research Laboratories** Oct. 2020 – Mar. 2021, Apr. 2021 – Dec. 2021
Remote Research Intern, Part-Time Researcher Massachusetts, USA
 - Mentor: Takaaki Hori, Niko Moritz, Jonathan Le Roux
 - Worked on semi-supervised speech recognition
 - Major publication(s): [Higuchi+, Interspeech’21] [Higuchi+, JSTSP’22]
- **Johns Hopkins University** Dec. 2019 – Mar. 2020
Visiting Scholar Maryland, USA
 - Mentor: Shinji Watanabe
 - Worked on non-autoregressive speech recognition
 - Major publication(s): [Higuchi+, Interspeech’20] [Higuchi+, ICASSP’21]
- **IBM Research AI** June 2019 – Oct. 2019
Research Intern Tokyo, Japan
 - Mentor: Masayuki Suzuki, Gakuto Kurata
 - Worked on speaker embeddings for speaker diarization
 - Major publication(s): [Higuchi+, ICASSP’20]
- **NTT Communication Science Laboratories** Aug. 2018 – Sept. 2018
Research Intern Kagawa, Japan
 - Mentor: Hirokazu Kameoka
 - Worked on voice conversion based on generative adversarial networks

EXTERNAL RESEARCH FUNDING

- **Super Global University (Visit to Carnegie Mellon University) | 1,000,000 JPY** *Mar. 2022 – June 2022*
from Top Global University Japan Project
 - In this study abroad program, I have participated in joint research with Prof. Shinji Watanabe at the Language Technologies Institute, Carnegie Mellon University. The primary goal of our project was to explore the integration of a pre-trained language model into end-to-end speech recognition.
 - Major publication(s): [Higuchi+, EMNLP Finding’22] [Higuchi+, ICASSP’23a] [Yan+, EACL’23]
- **ACT-X Frontier of Mathematics and Information Science | 4,500,000 JPY** *Oct. 2021 – Mar. 2024*
from Japan Science and Technology Agency (JST)
 - This research project aimed to improve the recognition accuracy and functionality of end-to-end speech recognition models, focusing on the development of technology for effectively extracting linguistic information from speech information. In particular, by explicitly designing a feature extraction process that incorporates linguistic knowledge, I have established a novel framework for end-to-end speech recognition that does not solely rely on data-driven approaches. Moreover, the proposed framework has been shown useful for tasks that require semantic understanding in conjunction with speech recognition.
 - Major publication(s): [Higuchi+, ICASSP’22b] [Higuchi+, ICASSP’23b] [Higuchi+, arXiv’23]
- **Research Fellowship for Young Scientists (DC1) | 2,200,000 JPY** *Apr. 2021 – Mar. 2024*
from Japan Society for the Promotion of Science (JSPS)
 - To enhance the practicality of speech-based communication interfaces, this research project aimed to develop a speech recognition system that operates with high accuracy and fast speed. To this end, I have constructed a non-autoregressive end-to-end speech recognition system based on masked language models. Compared to conventional autoregressive systems, the proposed model has been shown to achieve significant improvements in inference speed while maintaining a similar level of recognition accuracy.
 - Major publication(s): [Higuchi+, ICASSP’21] [Higuchi+, ASRU’21] [Zhao+, EUSIPCO’23]
- **Super Global University (Visit to Johns Hopkins University) | 1,000,000 JPY** *Dec. 2019 – Mar. 2020*
from Top Global University Japan Project
 - In this study abroad program, I have participated in joint research with Prof. Shinji Watanabe at the Center for Language and Speech Processing, Johns Hopkins University. The primary goal of our project was to explore an end-to-end speech recognition model architecture that can operate with fast inference speed.
 - Major publication(s): [Higuchi+, Interspeech’20] [Guo+, ICASSP’21] [Watanabe+, DSLW’21]

AWARDS

- **Best Reviewer Award** *Dec. 2023*
from IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)
- **ISS Young Researcher’s Award in Speech Field** *Sept. 2021*
from the Institute of Electronics, Information and Communication Engineers (IEICE)
- **Repayment Exemption for Graduate Students with Excellent Achievements (full-exemption)** *Mar. 2021*
from Japan Student Services Organization (JASSO)
- **Best Student Presentation Award** *Nov. 2020*
from the Acoustical Society of Japan (ASJ)
- **Yamashita SIG Research Award** *Aug. 2020*
from Information Processing Society of Japan (IPSJ)
- **Yahoo! JAPAN Award** *Jan. 2020*
from Information Processing Society of Japan (IPSJ) SIG-SLP

REVIEWING ACTIVITY

ACL: 2024 APSIPA: 2020, 2021 ASRU: 2023 ICASSP: 2023 Interspeech: 2023, 2024 SLT: 2022
SPECOM: 2024 TASLP: 2022, 2023, 2024

SKILLS

- **Language:** Japanese (native), English (fluent – TOEFL iBT: 102, TOEIC: 945)
- **Programming:** Python, C, C++, Java, JavaScript, LaTeX
- **Software & Tools:** PyTorch, Chainer, TensorFlow, ESPnet, Kaldi, Lingvo

PUBLICATIONS

Journal

1. Yosuke Higuchi, Niko Moritz, Jonathan Le Roux, Takaaki Hori, “**Momentum Pseudo-Labeling: Semi-Supervised ASR with Continuously Improving Pseudo-Labels**,” *IEEE Journal of Selected Topics in Signal Processing*, October 2022.

International Conference (First Author)

1. Yosuke Higuchi, Andrew Rosenberg, Yuan Wang, Murali Karthick Baskar, Bhuvana Ramabhadran, “**Mask-Conformer: Augmenting Conformer with Mask-Predict Decoder**,” *2023 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, December 2023.
2. Yosuke Higuchi, Tetsuji Ogawa, Tetsunori Kobayashi, Shinji Watanabe, “**BECTRA: Transducer-based End-to-End ASR with BERT-Enhanced Encoder**,” *2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, June 2023.
3. Yosuke Higuchi, Tetsuji Ogawa, Tetsunori Kobayashi, Shinji Watanabe, “**InterMPL: Momentum Pseudo-Labeling with Intermediate CTC Loss**,” *2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, June 2023.
4. Yosuke Higuchi, Brian Yan, Siddhant Arora, Tetsuji Ogawa, Tetsunori Kobayashi, Shinji Watanabe, “**BERT Meets CTC: New Formulation of End-to-End Speech Recognition with Pre-trained Masked Language Model**,” *Findings of the Association for Computational Linguistics: EMNLP 2022*, December 2022.
5. Yosuke Higuchi, Keita Karube, Tetsuji Ogawa, Tetsunori Kobayashi, “**Hierarchical Conditional End-to-End ASR with CTC and Multi-Granular Subword Units**,” *2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, May 2022.
6. Yosuke Higuchi, Niko Moritz, Jonathan Le Roux, Takaaki Hori, “**Advancing Momentum Pseudo-Labeling with Conformer and Initialization Strategy**,” *2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, May 2022.
7. Yosuke Higuchi, Nanxin Chen, Yuya Fujita, Hirofumi Inaguma, Tatsuya Komatsu, Jaesong Lee, Jumon Nozaki, Tianzi Wang, Shinji Watanabe, “**A Comparative Study on Non-Autoregressive Modelings for Speech-to-Text Generation**,” *2021 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, December 2021.
8. Yosuke Higuchi, Niko Moritz, Jonathan Le Roux, Takaaki Hori, “**Momentum Pseudo-Labeling for Semi-Supervised Speech Recognition**,” *22nd Annual Conference of International Speech Communication Association (Interspeech)*, August 2021.
9. Yosuke Higuchi, Shinji Watanabe, Hirofumi Inaguma, Tetsuji Ogawa, Tetsunori Kobayashi, “**Improved Mask-CTC for Non-Autoregressive End-to-End ASR**,” *2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, June 2021.
10. Yosuke Higuchi, Naohiro Tawara, Atsunori Ogawa, Tomoharu Iwata, Tetsunori Kobayashi, Tetsuji Ogawa, “**Noise-robust Attention Learning for End-to-End Speech Recognition**,” *28th European Signal Processing Conference (EUSIPCO)*, January 2021.
11. Yosuke Higuchi, Shinji Watanabe, Nanxin Chen, Tetsuji Ogawa, Tetsunori Kobayashi, “**Mask CTC: Non-Autoregressive End-to-End ASR with CTC and Mask Predict**,” *21st Annual Conference of International Speech Communication Association (Interspeech)*, October 2020.
12. Yosuke Higuchi, Masayuki Suzuki, Gakuto Kurata, “**Speaker Embeddings Incorporating Acoustic Conditions for Diarization**,” *2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, May 2020.
13. Yosuke Higuchi, Naohiro Tawara, Tetsunori Kobayashi, Tetsuji Ogawa, “**Speaker Adversarial Training of DPGMM-based Feature Extractor for Zero-Resource Languages**,” *20th Annual Conference of International Speech Communication Association (Interspeech)*, September 2019.

International Conference (Co-Author)

1. Tomoki Ariga, Yosuke Higuchi, Kazutoshi Hayasaka, Naoki Okamoto, and Tetsuji Ogawa, “**Parody Detection Using Source-Target Attention With Teacher-Forced Lyrics**,” *2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, April 2024.
2. Masao Someki, Nicholas Eng, Yosuke Higuchi, Shinji Watanabe, “**Segment-Level Vectorized Beam Search Based on Partially Autoregressive Inference**,” *2023 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, December 2023.
3. Tomoki Ariga, Yosuke Higuchi, Mitsunori Kanno, Rie Shigyo, Takato Mizuguchi, Naoki Okamoto, Tetsuji Ogawa, “**Spotting Parodies: Detecting Alignment Collapse Between Lyrics and Singing Voice**,” *31st European Signal Processing Conference (EUSIPCO)*, September 2023.
4. Huaibo Zhao, Yosuke Higuchi, Yusuke Kida, Tetsuji Ogawa, Tetsunori Kobayashi, “**Mask-CTC-based Encoder Pre-training for Streaming End-to-End Speech Recognition**,” *31st European Signal Processing Conference (EUSIPCO)*, September 2023.
5. Brian Yan, Siddharth Dalmia, Yosuke Higuchi, Graham Neubig, Florian Metze, Alan W Black, Shinji Watanabe, “**CTC Alignments Improve Autoregressive Translation**,” *17th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, May 2023.
6. Yifan Peng*, Siddhant Arora*, Yosuke Higuchi, Yushi Ueda, Sujay Kumar, Karthik Ganesan, Siddharth Dalmia, Xuankai Chang, Shinji Watanabe, “**A Study on the Integration of Pre-Trained SSL, ASR, LM and SLU Models for Spoken Language Understanding**,” *2022 IEEE Spoken Language Technology Workshop (SLT)*, January 2023.
7. Masao Someki, Yosuke Higuchi, Tomoki Hayashi, Shinji Watanabe, “**ESPnet-ONNX: Bridging a Gap Between Research and Production**,” *Asia-Pacific Signal and Information Processing Association Annual Summit and Conference 2022 (APSIPA)*, November 2022.
8. Keqi Deng*, Zehui Yang*, Shinji Watanabe, Yosuke Higuchi, Gaofeng Cheng, Pengyuan Zhang, “**Improving Non-Autoregressive End-to-End Speech Recognition with Pre-trained Acoustic and Language Models**,” *2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, May 2022.
9. Huaibo Zhao, Yosuke Higuchi, Tetsuji Ogawa, Tetsunori Kobayashi, “**An Investigation of Enhancing CTC Model for Triggered Attention-based Streaming ASR**,” *Asia-Pacific Signal and Information Processing Association Annual Summit and Conference 2021 (APSIPA)*, December 2021.
10. Shinji Watanabe, Florian Boyer, Xuankai Chang, Pengcheng Guo, Tomoki Hayashi, Yosuke Higuchi, Takaaki Hori, Wen-Chin Huang, Hirofumi Inaguma, Naoyuki Kamo, Shigeki Karita, Chenda Li, Jing Shi, Aswin Shanmugam Subramanian, Wangyou Zhang, “**The 2020 ESPnet Update: New Features, Broadened Applications, Performance Improvements, and Future Plans**,” *2021 IEEE Data Science & Learning Workshop (DSLW)*, June 2021.
11. Pengcheng Guo, Florian Boyer, Xuankai Chang, Tomoki Hayashi, Yosuke Higuchi, Hirofumi Inaguma, Naoyuki Kamo, Chenda Li, Daniel Garcia-Romero, Jiatong Shi, Jing Shi, Shinji Watanabe, Kun Wei, Wangyou Zhang, Yuekai Zhang, “**Recent Developments on ESPnet Toolkit Boosted by Conformer**,” *2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, June 2021.
12. Hirofumi Inaguma, Yosuke Higuchi, Kevin Duh, Tatsuya Kawahara, Shinji Watanabe, “**Orthros: Non-autoregressive End-to-end Speech Translation with Dual-decoder**,” *2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, June 2021.

Preprint

1. Yosuke Higuchi, Tetsuji Ogawa, Tetsunori Kobayashi, “**Harnessing the Zero-Shot Power of Instruction-Tuned Large Language Model in End-to-End Speech Recognition**,” *arXiv preprint arXiv:2309.10524*, September 2023.
 2. Hirofumi Inaguma, Yosuke Higuchi, Kevin Duh, Tatsuya Kawahara, Shinji Watanabe, “**Non-autoregressive End-to-end Speech Translation with Parallel Autoregressive Rescoring**,” *arXiv preprint arXiv:2109.04411*, September 2021.
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